

LI Databases and Web programming

Lab sheet Week 2

Scenario: To-Do Application

You are asked to design the database for a **To-Do Item Application**.

The application allows users to manage their **tasks** (to-do items), organize them by **projects**, and assign ownership. Each task also has a **priority**, a **status**, and can be linked to a **context**.

Here are the main details:

- A **To-Do Item** has:
 - a unique identifier,
 - a text description,
 - a due date, and
 - a completion date.
- Each **To-Do Item** belongs to one **Owner**.
 - An **Owner** has:
 - an ID,
 - a first name, and
 - a surname.
- Each **To-Do Item** is linked to a **Project**.
 - A **Project** has:
 - a project ID,
 - a project name, and
 - a budget.
- Each **To-Do Item** is associated with a **Context**.
 - A **Context** has:
 - a context ID, and
 - a description.
- Each **To-Do Item** also has a **Priority** (e.g., High, Medium, Low).
- Each **To-Do Item** also has a **Status** (e.g., Pending, In Progress, Completed).

Task 1: Extract Entities

From the scenario, list the **entities** and their **attributes**.

Write them in this format:

- Entity: Attributes

Task 3: Write the JDL File

After identifying the entities and relationships, write a **JDL model** for this system.

Task 4: Visualize the ER Diagram

- Open **JDL Studio** (<https://start.jhipster.tech/jdl-studio/>).
- Paste your JDL code.
- Generate and **draw the ER diagram**.